

# Luca Luchini

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## Summary

Mechanical Engineering student with hands-on experience in CAD, basic FEA, and engineering support in a production environment. Interested in design, product development, analysis, and testing, with additional exposure to circuits and electrical systems through coursework and independent learning. Brings a practical, fast-learning approach shaped by experience with precision assemblies, process improvement, and cross-functional team environments.

## Experience

### SUMMER INTERN | BGM ENGINEERING | JUNE 2025 – AUGUST 2025

- Designed a load-cell monitoring system to measure conformal-coat tank levels, improving maintenance efficiency, preventing downtime, and avoiding the need for a commercial system costing several thousand dollars.
- Supported production of precision assemblies with emphasis on quality, consistency, and efficiency.
- Streamlined assembly and test-line workflow by improving process steps and implementing practical efficiencies to increase throughput.
- Worked with engineers and production staff to support builds, troubleshoot issues, and maintain reliable output in a fast-paced manufacturing environment.
- Gained exposure to the product development process from design through final assembly, including design challenges, manufacturing constraints, and practical implementation.

### ELECTRICAL & POWER DELIVERY | WMU FORMULA SAE | SEPTEMBER 2024 – CURRENT

- Contributed to redesign of the rear differential mounting system by identifying failure points and refining geometry to improve structural reliability.
- Modeled updated differential mount components in SolidWorks and performed basic FEA to evaluate stress distribution and FOS.
- Performed basic FEA on drivetrain components to evaluate stress distribution and ensure parts could withstand expected torque and loading conditions
- Gained exposure to EV electrical systems, supporting team efforts and developing familiarity with drivetrain packaging, electrical integration, and system-level design considerations.

### 3D SCAN TECHNICIAN | BELFOR USA | MAY 2022 – CURRENT

- Provided 3D building scanning and modeling services using Matterport for a variety of large and small clients across the US
- Managed a small team of scanners and organized logistics for numerous jobs and trips across the US

### AUTO SHOP FOREMAN | OXFORD HIGH SCHOOL | SEPTEMBER 2019 – MAY 2021

- Worked across braking, electrical, suspension and drivetrain systems, developing practical understanding of mechanical components, wiring, diagnostics, and system interactions
- Learned about all automotive systems
- Diagnosed and serviced vehicles brought in by real clients
- Used hand and power tools, equipment, and instruments

## Education

### B.S. IN MECHANICAL ENGINEERING | OAKLAND UNIVERSITY, ROCHESTER, MI | JUNE 2028

- Minor in Mathematics & Business
- 3.5 Major GPA, 3.1 Overall GPA

## Skills

SolidWorks • AutoCAD • Mechanical Design • FEA Fundamentals • System Integration • Excel

KiCad • LTSpice • Circuits • C • Automotive Systems